MILITARY SPECIFICATION SHEET

CABLE ASSEMBLIES, RADIO FREQUENCY
(MADE FROM M17/28-RG58 CABLE)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the cable assemblies described herein shall consist of this document and the latest issue of Specification MIL-C-55427.

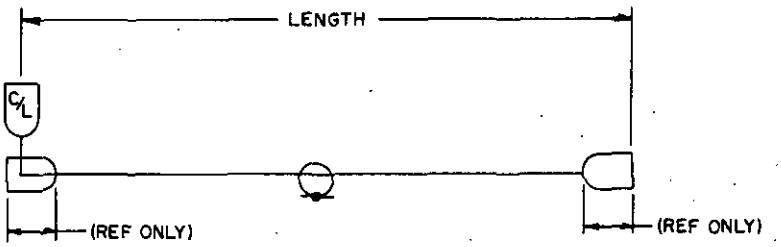


FIGURE 1. General configuration.

REQUIREMENTS:

Configuration: See figure 1.

Cable: Part No. M17/28-RG58 per MIL-C-17/28.

End terminations: See table I.

Length of assembly: As specified in part number.
Tolerance: +2%, -0%.

Dielectric withstanding voltage:
1500 volts rms at sea level - BNC and TNC series connectors.
1000 volts rms at sea level - SMA series connectors.

Thermal shock: MIL-STD-202, method 107, test condition A.

Flexure: 20 pounds minimum.

Cable retention:
Longitudinal pull - See table I.
Torque - 2 inch-pounds.

VSWR: 1:1.3 maximum at 0.5 to 1 GHz. See table I.

Insertion loss: 0.5 dB maximum at 1 GHz.

$$\left(\frac{\text{Length of cable}}{\text{Unit length}} \times \frac{\text{Loss}}{\text{Unit length}}\right) + \left(\text{Connector loss per table I}\right).$$

FSC 5995

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TABLE I. Cable assembly characteristics.

MIL-C-39012/ terminations	Cable retention	VSWR	Insertion loss
	Pounds (min)		dB (max)
16A	40	1.30 (50 mHz to 4 GHz)	.2 at 3 GHz
16B	60	↓	↓
16C	60		
16D	60		
17A	40		
17B	60		
17C	60		
17D	60		
20A	40	1.35 (50 mHz to 4 GHz)	.3 at 3 GHz
20B	60	↓	↓
20C	60		
20D	60		
26A	40	1.30 (50 mHz to 11 GHz)	.18 at 9 GHz
26B	60	↓	↓
26C	60		
26D	60		
27A	40		
27B	60		
27C	60		
27D	60		
30A	40		.21 at 9 GHz
30B	60		↓
30C	60		
30D	60		
55A	45	1.15 + 0.01 (F) GHz	.06 x $\sqrt{\text{Freq GHz}}$
55B	45	↓	↓
55C	45		
55D	45		
56A	45	1.15 + 0.02 (F) GHz	.15 x $\sqrt{\text{Freq GHz}}$
56B	45	↓	↓
56C	45		
56D	45		
57A	45	1.15 + 0.01 (F) GHz	.06 x $\sqrt{\text{Freq GHz}}$
57B	45	↓	↓
57C	45		
57D	45		

Part number: Consists of "M55427/01", two-letter code denoting end terminations (see table II), and cable length (expressed in centimeters by four digits).

Sample units for group C inspection: Minimum of two sample units per qualification group of table III (total of 12 sample units).

Group qualification: In accordance with table III.

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TABLE II. End termination codes and applicable data.

	Category				Termination		Shell material
	A	B 1/	C	D	MIL-C-39012/	Series	
Code - - - - -	A	B	C	D	16	BNC	Brass
Code - - - - -	E	F	G	H	17	BNC	Brass
Code - - - - -	J	K	L	M	20	BNC	Brass
Code - - - - -	N	P	Q	R	26	TNC	Brass
Code - - - - -	S	T	U	V	27	TNC	Brass
Code - - - - -	W	X	Y	Z	30	TNC	Brass
Code - - - - -	a	b	c	---	55	SMA	Corrosion resistant steel
Code - - - - -	d	e	f	---	56	SMA	Corrosion resistant steel
Code - - - - -	g	h	j	---	57	SMA	Corrosion resistant steel
Code - - - - -	---	k	---	---	55	SMA	Copper beryllium
Code - - - - -	---	l	---	---	56	SMA	Copper beryllium
Code - - - - -	---	m	---	---	57	SMA	Copper beryllium

1/ Not for use in airborne applications.

TABLE III. Group qualification.

Qualification group	Qualification of any one of the following terminations:		Qualifies all of the following connector terminations:	
	MIL-C-39012/	Category	MIL-C-39012/	Category
1	16 17 20 26 27 30	A	16 17 20 26 27 30	A
1/ 2	16 17 20 26 27 30	B	16 17 20 26 27 30	B
3	16 17 20 26 27 30	C or D	16 17 20 26 27 30	C or D
4	55 56 57	A	55 56 57	A
1/ 5	55 56 57	B	55 56 57	B
6	55 56 57	C or D	55 56 57	C or D

1/ Not for use in airborne applications.

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Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - AR, MI
Navy - SH, AS, OS
Air Force - 11, 17
DLA - ES

User activities:
Army - ME, AT, AV
Navy - MC
Air Force - 14, 19

Preparing activity:
Army - EL

Agent:
DLA - ES

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